

Proposal ID: 14**Title: Do NBA prediction markets exhibit systematic pricing bias correlated with media attention disparities?**

Your project proposal is well-structured and demonstrates a rigorous application of probability and hypothesis testing to the emerging field of prediction markets. Your research questions are clear and focused, centering on whether media attention creates a systematic divergence between market-implied probabilities and actual outcomes. With an anticipated sample of 1,000–3,000 games, your analysis will have strong statistical power. One of the major strengths of your approach is the use of Bootstrap Resampling to construct confidence intervals for pricing errors, allowing you to assess whether the “big market bias” represents a genuine effect or simply random variation. To further refine the analysis, you might consider using Hierarchical Indexing and GroupBy operations to stratify observations by levels of Media Disparity (e.g., High, Medium, Low) before conducting your hypothesis tests. This will let you compare mean pricing errors across groups directly, while staying within the methodological toolkit covered in class. Additionally, applying Gaussian Kernel Density Estimation (KDE) can help visualize differences in pricing-error distributions between high-media and low-media teams, making any systematic patterns easier to observe. Finally, ensure your data-preparation workflow includes a careful join between the Kalshi and Basketball-Reference datasets so you can incorporate Team Net Rating and compute a quality-adjusted pricing error using basic arithmetic operations. Overall, this is a well-constructed proposal that shows great potential for meaningful discovery.